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Detailed analytical model of a single-cylinder diesel engine in the crank angle domain

Y H Zweiri, J F Whidborne* and **L D Seneviratne**
Department of Mechanical Engineering, King's College London, UK

Abstract: A detailed analytical non-linear dynamic model for single-cylinder diesel engines is developed. The model describes the dynamic behaviour between fuelling and engine speed and includes models of the non-linear engine and dynamometer dynamics, the instantaneous friction terms and the engine thermodynamics. The model operates in the crank angle domain. The dynamometer model enables the study of the engine behaviour under loading. The instantaneous friction model takes into consideration the viscosity variations with temperature. Inertia variations with piston pin offset are presented. In-cycle calculations are performed at each crank angle, and the correct crank angles of ignition, speed variations, fuel supply and air as well as fuel burning rate are predicted. The model treats the cylinder strokes and the manifolds as thermodynamic control volumes by using the filling and emptying method. The model is validated using experimentally measured cylinder pressure and engine instantaneous speeds, under transient operating conditions, and gives good agreement. The model can be used as an engine simulator to aid diesel engines control system design and fault diagnostics.

Keywords: diesel engine modelling, simulation, dynamic modelling, thermodynamic modelling, transient response, fuelling control, engine diagnostics

NOTATION

a_i	reduction coefficients for the piston rings
A	heat transfer area (m^2)
A_m	exhaust manifold area (m^2)
A_p	piston area (m^2)
A_v	valve area (m^2)
b_1, b_2	length ratios
B_i	the width of the ring in the direction of motion (slider width) (m)
c	compression ratio
c_r	radial clearance (m)
C_d	discharge coefficient
C_p	specific heat
d	cylinder bore diameter (m)
d_i	wire diameter of the intake or exhaust valve spring (m)
d_o	mean coil diameter of the intake or exhaust valve spring (m)
d_r	piston ring diameter (m)
D	damping coefficient (N m/rad s)
D_i	inner exhaust manifold diameter
D_o	outer exhaust manifold diameter
E_i	nominal modulus of elasticity of piston ring (N/m^2)
f	fuel–air ratio
f_s	stoichiometric fuel–air ratio
f_t	total fuel fraction
F	equivalence ratio
F_f	friction force (N)
g	gap closure of the piston ring (m)
$G(\theta_1)$	geometric function
h_0	stagnation enthalpy
h_f	forced heat transfer convective coefficient
h_{for}	enthalpy of formation of the hydrocarbon
h_n	natural heat transfer convective coefficient
ii	subscript to identify each piston ring
ID	ignition delay
j	subscript to identify surfaces with different rates of heat transfer
J	moment of inertia of crankshaft, flywheel, main gear and rotating part of connecting rod (kg m^2)
J_c	moment of inertia of crankshaft (kg m^2)
J_R	moment of inertia of connecting rod (kg m^2)
J_1	moment of inertia of dynamometer rotating parts (kg m^2)
K	thermal conductivity

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* *Corresponding author: Department of Mechanical Engineering, King's College London, Strand, London WC2R 2LS, UK.*

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L	connecting rod length (m)
L_b	bearing length (m)
L_s	skirt length (m)
m_a	air mass
m_c	crankshaft mass
m_f	fuel mass
m_p	piston mass
m_r	connecting rod mass
$\acute{m}$	mass flowrate
$\acute{m}_{bf}$	burned fuel mass rate
$\acute{m}_f$	fuel rate
M	piston, rings, pin and small end of connecting rod mass (kg)
N_v	number of intake and exhaust valves
O_c	oil clearance (m)
p_{atm}	atmospheric pressure = 101 kPa
p_i	indicated pressure (Pa)
P	pressure
P_d	downstream pressure
P_u	upstream pressure
Q_{ht}	heat transfer
r	crank radius (m)
r_b	bearing radius (m)
R	gas constant
S	coupling stiffness
t	time (s)
T	temperature (K)
T_D	damping torque (N m)
T_{f1}	piston ring assembly friction torque (N m)
T_{f2}	crankshaft bearing friction torque (N m)
T_{f3}	valve train friction torque (N m)
T_{f4}	pumping losses torque (N m)
T_{f5}	pumps friction torque (N m)
T_i	indicated torque (N m)
T_{Lj}	engine load torque (N m)
T_m	exhaust manifold temperature (K)
T_r	reciprocating inertia torque (N m)
T_{rad}	apparent radiant temperature (K)
T_S	torsional stiffness torque (N m)
T_w	cylinder wall temperature (K)
u	specific internal energy
V_d	displacement volume (m^3)
V_l	valve lift (m)
y	piston displacement (m)
z	hydrodynamic friction coefficient (m)
β	connecting rod angle (rad)
δ	piston pin offset (m)
ϵ	bearing eccentricity (m)
ε	apparent grey body emissivity
η_{bm}	mixed friction coefficient
θ_1	crankshaft angular position (rad)
θ_2	dynamometer angular position (rad)
$\lambda(\theta_1)$	geometric function
μ	dynamic viscosity of the oil ($N\,s/m^2$)
ν	kinematic viscosity
ϖ	average exhaust flowrate
ρ	density (kg/m^3)
ς	volumetric coefficient
τ	moment of inertia of crankshaft assembly and the reciprocating parts ($kg\,m^2$)
ϕ	connecting rod angle when piston is at top dead centre (rad)
ω_1	engine speed (rad/s)
ω_2	coupling speed (rad/s)

In this paper, the derivative of some variable, x, with respect to time is denoted by $\dot{x}$ and the derivative with respect to crank angle by $\acute{x}$.

1 INTRODUCTION

The modelling of diesel engines for engine simulation and design has received much research attention. In addition, with the recent rapid advance of microprocessor technology, it is feasible to use these models for real-time applications, such as state estimation, control and fault diagnostics. However, there needs to be further research in modelling techniques, to improve predictive accuracy and to achieve more precise estimation and control.

Early studies into diesel engine modelling relied on the use of empirical data to develop linear dynamic models [**1**–**3**]. These models link steady state experimental data representing engine thermodynamics and gas flow with simple dynamic models of the mechanical components. The major disadvantage of such quasi-linear models is their heavy reliance on experimental data. Furthermore, they poorly represent the transient response of the engine. More recently, non-linear engine simulation models using the filling and emptying method and the method of characteristics have been developed [**4**–**7**] for the purpose of engine design and performance predictions.

A number of models have been developed for the purpose of diesel engine control and estimation. In reference [**8**], a mean value model was used. The model consists of several empirical algebraic and first-order differential equations in order to achieve real-time implementation. The main drawback of this type of model is the requirement of empirical test data to fit model equations. Diesel engine models for electronic control were developed in references [**9**] and [**10**]. The in-cylinder combustion process was included by using a single-zone model and a filling and emptying modelling technique. In reference [**9**], several modifications to the model are proposed to cut down the computational time. These models do not predict in-cycle crankshaft angular speed variations. In reference [**11**] hybrids of two engine models, a mean torque production model and a simplified cylinder-by-cylinder model, are investigated in order to provide control engineers with tools for developing control and diagnostic algorithms. The implementation of the mean torque model requires empirical test data to fit the model

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equations, and the simplified cylinder-by-cylinder model is not able to predict accurately the in-cycle variations of the engine states.

Non-linear transient simulations of the diesel engine in order to predict engine torque and speed have been carried out in both references [**12**] and [**13**]. Engine control schemes generally operate on engine events which are periodic in the crank angle domain rather than on a real-time clock. Hence, for the purposes of model-based control and fault diagnostic schemes, it is preferable that the engine state variables are based in the crank angle domain rather than the time domain. In both references [**12**] and [**13**], the simulations are performed in the time domain. In addition, the inertia variation is not included in either model. Furthermore, the instantaneous friction components model included in reference [**13**] does not take into consideration the viscosity variation with temperature, and the empirical coefficients of the friction components are required to be calibrated for each type of engine.

Some investigators have studied engine dynamic modelling, describing the relationship between the net engine torque and the angular speed of the crankshaft [**14–17**]. None of these studies includes the inertia variations of the crankshaft assembly and the piston pin offset.

In this paper a combined dynamic, instantaneous friction and thermodynamic engine model is developed and used to predict the in-cycle variations of the engine states. To enable the study of the engine behaviour under loading, a dynamic dynamometer model is also included. Engine events for control and estimation are generally periodic with respect to crank angle rather than time, and hence the engine model is totally based in the crank angle domain. The proposed model does not require any empirical inputs and takes into consideration the inertia variations of the crankshaft assembly with piston pin offset, and the friction model takes the viscosity variation with temperature. Some details of the model development are provided in the Appendix. The model treats the cylinder strokes and the manifolds as thermodynamic control volumes by using the filling and emptying method, solving energy and mass conservation equations with submodels for combustion and heat transfer. The model has been implemented in MATLAB/SIMULINK [**18, 19**] and in-cycle calculations are performed at each crank angle.

A general engine model will consist of both dynamic and thermodynamic equations. The dynamic equations are developed in the next section. The model inputs are the air and fuel flowrates and the load on the engine. Then, given the defined engine parameters, the model should predict the crankshaft speed variations and other engine states. The non-linear dynamic model in reference [**20**], including an instantaneous friction components model [**21**], is improved by computing inertia variations of the crankshaft assembly, and the time domain is transformed to the crank angle domain.

The thermodynamic model presented in Section 3 is based on the filling and emptying method. Three thermodynamic control volumes are considered: the inlet manifold, the cylinder and the exhaust manifold. A schematic diagram of the engine is shown in Fig. 2. Equations for the conservation of mass and energy, gas property relations and perfect gas law are solved in the crank angle domain. A single-zone combustion model is used for simulating the theoretical fuel burning rate. The cylinder is treated as one control volume with homogeneous temperature and pressure.

In Section 4 the implementation is described, followed by some simulation results to show the model behaviour and validation. Finally, there is a discussion and some conclusions are drawn.

2 ENGINE AND DYNAMOMETER DYNAMIC MODEL

Figure 1 shows a model of an engine coupled to a dynamometer. The model is discussed in the time domain in references [**20**] and [**22**].

To develop an engine dynamic model in the crank angle domain, consider the crankshaft angle θ_1 as an independent variable. Then the independent variable transformation can be obtained from reference [**23**] as follows:

$$\dot{\theta}_1 = \frac{\mathrm{d}\theta_1}{\mathrm{d}t} = \omega_1(\theta_1) \tag{1}$$

$$\ddot{\theta}_1 = \omega_1 \frac{\mathrm{d}\omega_1}{\mathrm{d}\theta_1} \tag{2}$$

$$\dot{\theta}_2 = \omega_1 \frac{\mathrm{d}\theta_2}{\mathrm{d}\theta_1} = \omega_2(\theta_1) \tag{3}$$

$$\ddot{\theta}_2 = \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = \omega_1 \frac{\mathrm{d}\omega_2}{\mathrm{d}\theta_1} \tag{4}$$

where t is time, ω_1 is the angular velocity of the engine crankshaft and ω_2 is the angular velocity of the dynamometer coupling. From equation (3), the relationship between the dynamometer angular position θ_2 and the independent variable θ_1 is given by

$$\int_0^{\theta_2} \mathrm{d}\theta_2 = \int_0^{\theta_1} \frac{\omega_2(\theta_1)}{\omega_1(\theta_1)} \mathrm{d}\theta_1 \Rightarrow \theta_2 = \int_0^{\theta_1} \frac{\omega_2(\theta_1)}{\omega_1(\theta_1)} \mathrm{d}\theta_1 \tag{5}$$

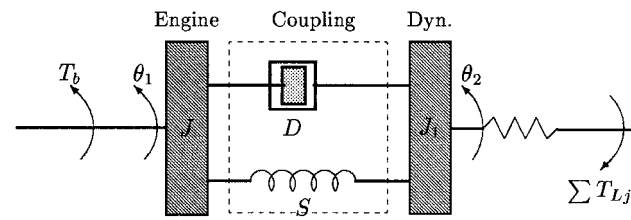

Fig. 1 An engine coupled to a dynamometer

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The following two equations, derived from Lagrangian principles, describe the dynamics of the system in the crank angle domain:

$$\dot{\omega}_1 = \frac{1}{\tau\omega_1}\left\{T_i(\theta_1) - \left[MrG(\theta_1)G_1(\theta_1) + \frac{1}{2}\frac{\partial J(\theta_1)}{\partial\theta_1}\right]\omega_1^2 - \sum_{k=1}^{5} T_{fk} - T_S - T_D\right\} \quad (6)$$

$$\dot{\omega}_2 = \frac{1}{J_1\omega_1}(T_D + T_S - T_L) \quad (7)$$

where the moment of inertia of crankshaft assembly and the reciprocating parts, τ, is given by $\tau = J(\theta_1) + MrG(\theta_1)G_2(\theta_1)$.

The indicated engine torque, T_i, is generated by the conversion of energy from chemical to thermal to mechanical, during the combustion process. The relationship between the indicated gas pressure, P_i, and the indicated torque, T_i, is deterministic and is a function of engine geometry. This relationship is given by

$$T_i = (P_i - P_{atm})A_p rG(\theta_1) \quad (8)$$

where

$$G(\theta_1) = \frac{\sin(\theta_1 + \beta)}{\cos\beta} = \sin\theta_1 + \sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 \quad (9)$$

and $\lambda(\theta_1)$ is a geometric function given by

$$\lambda(\theta_1) = 1 - \left[\frac{\delta + r\sin(\theta_1 - \phi)}{L}\right]^2 \quad (10)$$

where β is the angle of connecting rod, r is the crank radius (equal to half of the stroke), L is the connecting rod length, δ is the piston pin offset and ϕ is the connecting rod angle when the piston is at the top dead centre position.

From the piston–crank geometry, the piston displacement, y, is given by

$$y = \sqrt{(r+L)^2 - \delta^2} - [L\cos\beta + r\cos(\theta_1 - \phi)] \quad (11)$$

where the angles ϕ and β are given by

$$\phi = \sin^{-1}\frac{\delta}{r+L} \quad \text{and} \quad \beta = \sin^{-1}\frac{\delta + r\sin(\theta_1 - \phi)}{L} \quad (12)$$

The reciprocating torque, T_r, is produced by the motion of the piston assembly and the small end of the connecting rod and is given by

$$T_r = MrG(\theta_1)\ddot{y} = MrG(\theta_1)[G_1(\theta_1)\dot{\theta}_1^2 + G_2(\theta_1)\ddot{\theta}_1] \quad (13)$$

where $G_1(\theta_1)$ and $G_2(\theta_1)$ are functions of the engine geometry:

$$G_1(\theta_1) = r\left\{\cos(\theta_1 - \phi)\left[1 + \frac{(r/L)\cos(\theta_1 - \phi)}{\sqrt{\lambda(\theta_1)^3}}\right] - \sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\sin(\theta_1 - \phi)\right\} \quad (14)$$

$$G_2(\theta_1) = r\left[\sin(\theta_1 - \phi) + \sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos(\theta_1 - \phi)\right] = rG(\theta_1 - \phi) \quad (15)$$

M is the mass of piston, rings, pin and the small end of the connecting rod and $\ddot{y}$ is the acceleration of the reciprocating components.

The torsional stiffness torque, T_S, and damping torque, T_D, at the coupling between the engine and the dynamometer, are given by

$$T_S = S(\theta_1 - \theta_2) \quad (16)$$

and

$$T_D = D(\dot{\theta}_1 - \dot{\theta}_2) \quad (17)$$

where S and D are stiffness and damping coefficients respectively of the coupling between the engine and dynamometer.

2.1 Friction model

The friction torque terms include piston assembly friction, T_{f1}, bearing friction torque, T_{f2}, valve train friction torque, T_{f3}, pumping losses torque, T_{f4}, and pumps losses torque, T_{f5} [**21**]. These are subtracted from the instantaneous indicated torque value to produce the brake torque at the shaft. Subsequently, the resistance torque, $\sum_{j=1}^{n} T_{Lj}$, which is the result of external loading imposed on the engine by the dynamometer, is subtracted from the brake torque and the net value is passed on to the engine dynamic model.

2.1.1 Ring friction torques

It has been observed that the ring assembly lubrication mode is hydrodynamic except near the top and bottom dead centre positions where the oil film breaks down. The ring assembly friction torques can be found by assuming that the friction force is equal to the product of the normal load between the ring assembly and the liner, and the friction coefficient. Thus the ring assembly friction torque can be expressed as

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$$T_{f1} = \eta r|G(\theta_1)| \left\{ \sum_{ii=1}^{N} \left[\frac{E_{ii} g}{7.07 d_r (d_r/B_{ii} - 1)^3} \right] \pi d_r B_{ii} + \sum_{ii=1}^{N} a_{ii} |p_i - p_{atm}| \pi d_r B_{ii} + \frac{|p_i - p_{atm}|(\pi/4) d^2 - M G_1(\theta_1) \dot{\theta}_1^2}{\eta + G_3(\theta_1)} \right\} \tag{18}$$

where

$$G_3(\theta_1) = \frac{L\sqrt{1 - \{[\delta + r\sin(\theta_1 - \phi)]/L\}^2}}{\delta + r|\sin(\theta_1 - \phi)|} \tag{19}$$

The coefficient of friction for hydrodynamic lubrication, η, is directly proportional to the piston speed and oil viscosity and inversely proportional to the ring load. Further details can be found in reference [**21**]. Thus

$$\eta = \begin{cases} c_1 - (c_1 - z)|\sin\theta_1|, & \text{for } 1.5\pi \leqslant \theta_1 \leqslant 2.5\pi \\ z, & \text{otherwise} \end{cases} \tag{20}$$

where z is the hydrodynamic friction coefficient given by

$$z = \sqrt{\frac{\mu \dot{\theta}_1 r |G(\theta_1)|}{L_r}} \tag{21}$$

and L_r is the load per unit length given by

$$L_r = \sum_{ii=1}^{N} \left[\frac{E_{ii} g}{7.07 d_r (d_r/B_{ii} - 1)^3} + |p_i - p_{atm}| \right] B_{ii} \tag{22}$$

2.1.2 *Skirt friction torque*

In reciprocating internal combustion engines, the piston skirt profile is designed to prevent local contact due to thermal expansion and to reduce friction losses and piston slap, by ensuring that a proper lubricating oil film thickness for maximum wedge effect and an appropriate piston pin offset are maintained.

The piston skirt friction is derived by applying Newton's law for viscous friction and can be expressed as

$$T_{f2} = \left[\frac{\mu \dot{\theta}_1 r G(\theta_1)}{O_c} \right] d L_s r G(\theta_1) \tag{23}$$

2.1.3 *Bearings friction torque*

The main journal bearings are sited at the crank, both ends of the rod and the camshaft. These can be analysed using the Reynolds approximation to the Navier–Stokes equations. The bearings are treated as axisymmetric and two dimensional to account for their finite length. The bearings generally operate in the hydrodynamic lubrication mode except around top dead centre. By solving the Reynolds equation, the instantaneous friction torque during hydrodynamic lubrication mode (for $0 \leqslant \theta_1 < 1.9\pi$ and $2.4\pi < \theta_1 \leqslant 4\pi$) on the journal bearing is given by

$$T_{f3} = \frac{F_t \epsilon}{2} + \frac{2\pi \mu \dot{\theta}_1 r_b^3 L_b G_4(\theta_1)}{\sqrt{(c_r^2 - \epsilon^2)}} \tag{24}$$

where the geometrical function $G_4(\theta_1)$ is given as

$$G_4(\theta_1) = 1 + \frac{\cos\theta_1}{\sqrt{(L/r)^2 - \sin\theta_1^2}} \tag{25}$$

Because of the abrupt increase of friction torque shortly after the top dead centre, the mode of the lubrication is mixed and the friction coefficient, η_{bm}, depends on the properties of the pair of rubbing materials. η_{bm} has a negative slope in the Stribeck diagram and is given by

$$\eta_{bm} = c_2 \left(\frac{\mu \dot{\theta}_1}{|p_i - p_{atm}|} \right)^{-n} \tag{26}$$

where the index number n is taken as $\frac{1}{3}$. Thus the proposed formula to calculate the bearing friction torque around the top dead centre, for $1.9\pi \leqslant \theta_1 \leqslant 2.4\pi$, is

$$T_{f4} = \eta_{bm} r_b \times \left\{ \frac{(\pi/4) d^2 |p_i - p_{atm}| - r\dot{\theta}_1^2 M[\cos\theta_1 + (r/L)\cos 2\theta_1]}{G_5(\theta_1)} \times |\cos(-2\theta_1 + 4.3\pi)| \right\} \tag{27}$$

where $G_5(\theta_1)$ is given by

$$G_5(\theta_1) = \sqrt{1 - \left[\frac{\delta + r\sin(\theta_1 - \phi)}{L} \right]^2} \tag{28}$$

and $|\cos(-2\theta_1 + 4.3\pi)|$ is an empirical term to take into account the behaviour of the mixed friction torque around top dead centre.

2.1.4 *Valve train friction torque*

The valve train carries high loads over the entire speed range of the engine. Loads acting on the valve train at lower speeds are primarily due to the spring forces, while at higher speeds the inertia forces of the component masses dominate. Valve train friction accounts for the friction in cam/follower, rocker arm/pivot shaft, camshaft bearings, valve/valveguides and seals. To simplify the analysis, the valve train is considered as a system represented by the cam/follower and is assumed to operate in the boundary lubrication mode.

The valve train friction torque is formulated to depend on the load and the engine speed and is given by

$$T_{f5} = \left(\frac{d_i^4 G_v}{8 d_o^3 N_t} \right) V_1 N_v (1 - c_3 \dot{\theta}_1) r |G(\theta_1)| \tag{29}$$

2.1.5 *Auxiliaries and gears torque losses*

This component models the friction due to the unloaded journal bearings, water pump, fuel pump, oil pump,

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generator and gears. A hydrodynamic lubrication mode is assumed, and the friction coefficient is calculated as in equation (21), with unit oil-film pressure. If the terms which represent the normal load and the effective radius of the auxiliaries are assumed to be unity, then the friction torque for auxiliaries is formulated as

$$T_{f6} = \sqrt{\mu \dot{\theta}_1} \tag{30}$$

2.1.6 Viscosity variations with temperature

The variation of temperature and hence oil viscosity is an important factor in the engine friction calculations. The viscosity of oil decreases with increasing temperature. This trend is typical for Newtonian regimes for oils, where the viscosity is independent of the shear rate. Furthermore, the viscosity of oils increases with pressure. At certain times during hydrodynamic lubrication, the loads increase the oil pressure, which in turn increases the viscosity. The oil viscosity as a function of temperature and pressure for the SAE 30 oil is determined by a non-linear least-squares method for exponential function and is given by

$$\mu = 7.849 \times 10^{-5} \times \exp\left(\frac{-8.670 \times 10^{-3} T^2 - 1.15311 T + 1361}{T + 133} + \frac{p}{105}\right) \tag{31}$$

In this study, only the effects of temperature variations are taken into consideration, because oil pressure has only a minor effect on the engine oil viscosity, and hence $p = p_{atm}$ is used in equation (31).

2.2 Inertia variation

Reciprocating mechanisms have variable inertia owing to the change of geometry through a crank revolution. The piston and connecting rod masses change their position relative to the crankshaft axis and hence change the effective inertia about this axis. The same path is followed by these parts during each revolution of the crankshaft and thus the inertia variation is smooth and periodic. A variable inertia (with respect to the crankshaft rotational axis) as a function of the crankshaft position is used. For a given crank angle, the inertia with respect to the centre line of the crankshaft is defined by considering the equivalent inertia of the piston, connecting rod and crankshaft assembly [**24**–**27**]. In this study the piston pin offset is taken into consideration during the analysis of the variable inertia.

The crank angle varying inertia of an engine crankshaft assembly is derived in the Appendix and is given by

$$J(\theta_1) = J_c + m_c(b_2 r)^2 + J_R\left[\left(\frac{r}{L}\right)^2 \frac{1}{\lambda(\theta_1)}\cos^2\theta_1\right] + m_p r^2\left[\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]^2 + m_R r^2(1-b_1)^2\cos^2\theta_1 + m_R r^2\left[b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]^2 \tag{32}$$

and

$$\begin{aligned}\frac{\partial J(\theta_1)}{\partial \theta_1} = {} & 2J_R\left[\left(\frac{r}{L}\right)^3 \frac{\sqrt{1-\lambda(\theta_1)}}{\lambda(\theta_1)^2}\cos^3\theta_1 - \left(\frac{r}{L}\right)^2 \frac{1}{\lambda(\theta_1)}\cos\theta_1\sin\theta_1\right] - 2m_R(1-b_1)^2 r^2\cos\theta_1\sin\theta_1 \\ & - m_R r^2\left[b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 - \sin\theta_1\right]\left\{-\left[\frac{r b_1}{L\sqrt{\lambda(\theta_1)^3}}\right]\cos^2\theta_1 - \cos\theta_1 + b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\sin\theta_1\right\} \\ & - m_p r^2\left[\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]\left\{\left[\frac{r}{L\sqrt{\lambda(\theta_1)^3}}\right]\cos^2\theta_1 - \cos\theta_1 + \sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\sin\theta_1\right\}\end{aligned} \tag{33}$$

where J_c is the moment of inertia of the crankshaft, J_R is the moment of inertia of the connecting rod, m_c is the mass of the crankshaft, m_R is the mass of the connecting rod, m_p is the mass of the piston, b_1 is the length ratio of CB to CA and b_2 is the length ratio of ED to DC as shown in Fig. 2.

2.3 Cylinder volume and area

From the piston–crank mechanism, the cylinder volume is

$$V = \frac{V_d}{c-1} + \frac{\pi d^2}{4}\left[\sqrt{(r+L)^2 - \delta^2} - \left\{\sqrt{L^2 - [\delta + r\sin(\theta_1 - \phi)]^2} + r\cos(\theta_1 - \phi)\right\}\right] \tag{34}$$

The cylinder heat transfer area is given by

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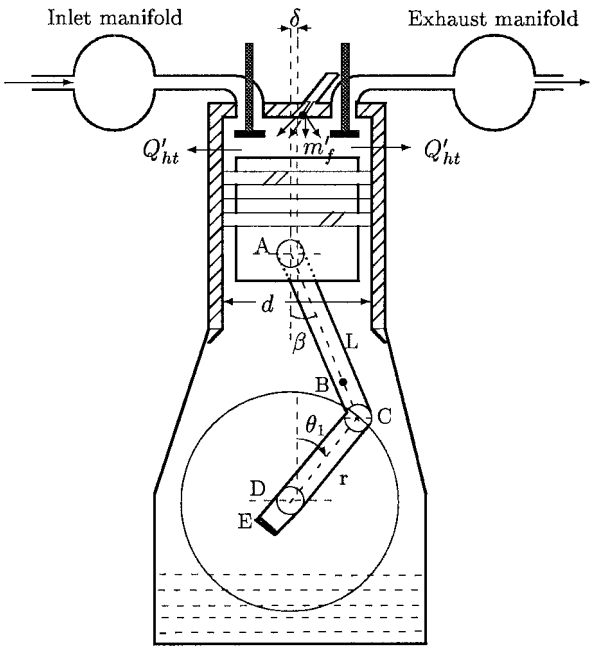

Fig. 2 Single-cylinder diesel engine

$$A = \alpha\frac{\pi d^2}{4} + \pi d\left[\sqrt{(r+L)^2-\delta^2} - \left\{\sqrt{L^2-[\delta + r\sin(\theta_1-\phi)]^2} + r\cos(\theta_1-\phi)\right\}\right] \quad (35)$$

where $\alpha > 2$ for a non-flat piston and cylinder head and $\alpha = 2$ for a flat piston top and cylinder head bottom. The derivative of equation (34) with respect to crank angle is

$$\acute{V} = \frac{\pi d^2}{4} r\left\{\sin(\theta_1-\phi) + \frac{\cos(\theta_1-\phi)[\delta + r\sin(\theta_1-\phi)]}{\sqrt{L^2-[\delta + r\sin(\theta_1-\phi)]^2}}\right\} \quad (36)$$

3 THERMODYNAMIC MODELLING

3.1 Conservation of mass and energy

The filling and emptying method treats the cylinder and each manifold as a thermodynamic control volume. Using the first law of thermodynamics for an open system and the ideal gas law, the temperature differential equation for each control volume can be written as [**4**, **9**]

$$\acute{T} = \frac{\left(\Sigma_j \acute{Q}_{htj} + \Sigma_{in} h_{0in}\acute{m}_{in} - \Sigma_{out} h_{0out}\acute{m}_{out} + \acute{m}_{bf}h_{for} - u\acute{m}\right)(1/m) - (RT/V)\acute{V} - (\partial u/\partial F)\dot{F}}{\partial u/\partial T} \quad (37)$$

where subscript j denotes surfaces with different rates of heat transfer, subscripts in and out denote the input to and output from the control respectively, F is the equivalence ratio, m_{bf} is the burned fuel mass, Q_{ht} is the heat transfer, R is the gas constant, h_0 is the stagnation enthalpy, hence implicitly accounting for kinetic energy, h_{for} is the enthalpy of formation of the hydrocarbon, u is the specific internal energy, and P, T, V and m are the control volume pressure, temperature, volume and mass respectively.

The specific internal energy u is assumed to be a function of temperature T and the equivalence ratio F only. Since dissociation effects are small [**9**], the influence of pressure is neglected. A curve fit of combustion product data as a continuous function is used [**4**], based on absolute zero temperature of the elements. Thus

$$u = u(T, F) \quad (38)$$

The gas constant R is given by

$$R = \frac{0.278 + 0.02F}{1 + f_s F} \quad (39)$$

where f_s is the stoichiometric fuel–air ratio.

Mass conservation for both air and fuel is required. Considering the total mass (fuel and air) gives

$$\acute{m} = \sum_{in}\acute{m} - \sum_{out}\acute{m} + \sum\acute{m}_{bf} \quad (40)$$

The total mass is given by

$$m = m_a + m_{bf} \quad (41)$$

where m_a is the mass of the air and m_{bf} is the mass of the burnt fuel. The fuel–air equivalence ratio is defined as

$$F = \frac{f}{f_s} \quad (42)$$

where f is the fuel–air ratio. The term $\acute{F}$ in equation (37) can be found in reference [**9**]:

$$\acute{F} = \left(\frac{1 + f_s F}{m}\right)\left(\frac{1 + f_s F}{f_s}\acute{m}_{bf} - F\acute{m}\right) \quad (43)$$

3.2 Ports and valves mass flowrates

A one-dimensional model for flow through a valve (or port), using the analogy of an orifice having an equivalent area, is used. Applying the energy equation from upstream to downstream for isentropic steady flow, and assuming that the inlet velocity is negligible, for subsonic flow $P_d/P_u > [2\gamma/(\gamma+1)]^{\gamma/(\gamma-1)}$, the mass flowrate has the form

$$\acute{m} = C_d A_v P_u\sqrt{\frac{2\gamma}{\gamma-1}\frac{1}{RT_u}\left[\left(\frac{P_d}{P_u}\right)^{2/\gamma} - \left(\frac{P_d}{P_u}\right)^{(\gamma+1)/\gamma}\right]} \quad (44)$$

where C_d is the discharge coefficient, γ is the specific heat ratio and A_v is the valve or port area.

For sonic flow $P_d/P_u \leqslant [2\gamma/(\gamma+1)]^{\gamma/(\gamma-1)}$, the mass

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flowrate has the form

$$\acute{m} = C_d A_v P_u \sqrt{\frac{\gamma}{RT}\left(\frac{2}{\gamma+1}\right)^{(\gamma+1)/(\gamma-1)}} \tag{45}$$

3.3 Combustion

Many combustion models have been discussed in references [**4**], [**5**] and [**28**]. The single-zone model proposed in reference [**9**] is used here, because it is widely accepted and can be used to predict both premixed and diffusion burning. The ignition delay (ID) equations as a function of cylinder pressure and temperature can be used to calculate the start of the combustion [**5**]:

$$\mathrm{ID} = 3.45\left(\frac{P}{101.3}\right)^{-1.02} \mathrm{e}^{2100/T} \tag{46}$$

with the constraint

$$\int_{\theta_{1\mathrm{inj}}}^{\theta_{1\mathrm{ign}}} \frac{\mathrm{d}\theta_1}{\mathrm{ID}} = 1 \tag{47}$$

where $\theta_{1\mathrm{inj}}$ is the angle at injection and $\theta_{1\mathrm{ign}}$ is the angle at ignition. The normalized premixed burning rate is given by

$$\acute{m}_{\mathrm{fpre}} = k_{p1} k_{p2} \theta_{1\mathrm{norm}}^{k_{p1}-1} (1 - \theta_{1\mathrm{norm}}^{k_{p1}})^{k_{p2}-1} \tag{48}$$

while the normalized diffusion burning rate is given by

$$\acute{m}_{\mathrm{fdiff}} = k_{d1} k_{d2} \theta_{1\mathrm{norm}}^{k_{d1}-1} \exp(-k_{d1} \theta_{1\mathrm{norm}}^{k_{d2}}) \tag{49}$$

where $\theta_{1\mathrm{norm}}$ is a non-dimensional angle, increasing from 0 at start of combustion to 1 at the end of combustion. The parameters k_{p1}, k_{p2}, k_{d1} and k_{d2} have been correlated with fundamental factors having the strongest influence on the combustion rate. The best-fit values of these parameters for a range of direct injection diesel engines were presented in reference [**9**].

The fraction of the total fuel that goes to the premixed burning, f_t, is given by [**9**]

$$f_t = 1 - 0.926 F^{0.37}\, \mathrm{ID}^{-0.26} \tag{50}$$

Assuming complete combustion in 125° of crank angle, the normalized angle is defined as

$$\theta_{1\mathrm{norm}} = \frac{\theta_1 - \theta_{1\mathrm{ign}}}{125} \tag{51}$$

Finally, the fuel burning rate is given by [**9**]

$$\acute{m}_{\mathrm{fnorm}} = f_t \acute{m}_{\mathrm{fpre}} + (1 - f_t)\acute{m}_{\mathrm{fdiff}} \tag{52}$$

$$\acute{m}_{\mathrm{bf}} = \frac{m_f \acute{m}_{\mathrm{fnorm}}}{\theta_{1\mathrm{com}}} \tag{53}$$

where $\theta_{1\mathrm{com}}$ is the combustion period in degrees.

3.4 Cylinder heat transfer

An instantaneous convective heat transfer is defined by

$$\acute{Q}_{\mathrm{ht1}} = h_{\mathrm{con}} A_v (T - T_{\mathrm{wall}}) \tag{54}$$

where h_{con} is the heat transfer coefficient based on forced convection and is given by [**29**]

$$h_{\mathrm{con}} = 130 V^{-0.06} P^{0.8} T^{-0.4} (N_p + 1.4)^{0.8} \tag{55}$$

where N_p is the mean piston speed and T_{wall} is the cylinder wall temperature. The primary sources of radiative heat transfer in diesel engines are the high temperature burned gases and soot particles. Following work in reference [**30**], the instantaneous radiant heat transfer can be expressed as

$$\acute{Q}_{\mathrm{ht2}} = \varepsilon \sigma A_v (T_{\mathrm{rad}}^4 - T_{\mathrm{wall}}^4) \tag{56}$$

where ε is the apparent grey body emissivity, σ is the Stephan–Boltzman constant and T_{rad} is the apparent radiant temperature. The adiabatic flame temperature, $T_{F=1.1}$, is the temperature of the hydrocarbon and air combustion products, when fuel–air ratio is slightly greater than stoichiometric ($F = 1.1$). The apparent radiant temperature T_{rad} is calculated as the mean of the adiabatic flame temperature and the average bulk gas temperature, i.e.

$$T_{\mathrm{rad}} = \frac{T + T_{F=1.1}}{2} \tag{57}$$

The temperature of combustion products at $F = 1.1$ is computed from a correlation of instantaneous air temperature and pressure as follows [**30**]:

$$T_{F=1.1} = [1 + 0.000231\,7(T - 950)] \times (2726.3 + 0.906P - 0.003233P^2) \tag{58}$$

for 800 K $< T <$ 1200 K, and

$$T_{F=1.1} = [1 + 0.000249(T - 650)](2487.3 + 4.7521P - 0.11065P^2 + 0.000898P^3) \tag{59}$$

for 450 K $< T <$ 800 K.

The apparent emissivity is assumed to vary linearly between its maximum value 0.9 and zero over the expansion stroke. The cylinder wall temperature is updated at each step by using the one-dimensional heat conduction model, and an electrical analogy model, shown in Fig. 3:

$$\acute{Q}_{\mathrm{ht1}} + \acute{Q}_{\mathrm{ht2}} = \frac{T_w - T_{\mathrm{coolant}}}{R_{wc} + R_w} \tag{60}$$

where $R_{wc} = 1/h_{\mathrm{coolant}} A$ is the thermal resistance from wall to coolant. The thermal resistance for conduction through the wall is $R_w = t/KA$, where t is the wall thickness and K is the thermal conductivity. A fourth-order

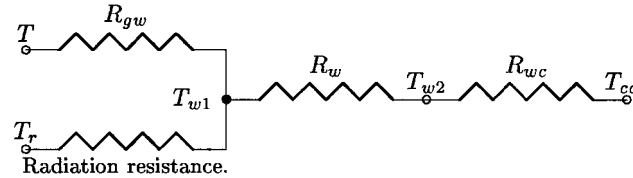

Fig. 3 Thermal resistance for the cylinder

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polynomial equation in T_m is obtained and the ith solution gives the wall temperature:

$$\frac{T - T_{wall}}{R_{gw}} + \varepsilon\sigma A(T_{rad}^4 - T_{wall}^4) = \frac{T_w - T_{coolant}}{R_{wc} + R_w} \quad (61)$$

where $R_{gw} = 1/h_{con}A$ is the thermal resistance from gas to wall. Thus the cylinder heat release rate can be calculated from equations (54) and (56):

$$\dot{Q}_{ht} = \dot{Q}_{ht1} + \dot{Q}_{ht2} \quad (62)$$

3.5 Exhaust manifold heat transfer

The exhaust gas flow is considered to be turbulent. The forced heat convection between the exhaust gas and the manifold wall, the natural convection and the radiation between the manifold and ambient air are taken into account. The conductive transfer through the wall thickness is neglected. An electrical analogy model is shown in Fig. 4. The manifold is assumed to have a cylindrical shape and homogeneous gas and wall temperatures. The heat transfer rate from exhaust gas to manifold is given by

$$\dot{Q}_{ht} = h_f A_m (T - T_m) \quad (63)$$

The heat transfer rate from manifold to air is given by

$$\dot{Q}_{ma} = h_n A_m (T_m - T_{air}) \quad (64)$$

The heat transfer rate due to radiation is given by

$$\dot{Q}_r = \varepsilon\sigma A_m (T_m^4 - T_{air}^4) \quad (65)$$

h_f is the forced heat convective coefficient given by [**31**]

$$h_f = 0.023 \frac{K}{D_i} R_e^{0.8} P_r^{0.3} \quad (66)$$

where R_e is the Reynolds number and P_r is the Prandtl number:

$$R_e = \frac{\varpi D_i}{\nu} \quad (67)$$

$$P_r = \frac{C_p \mu}{K} \quad (68)$$

where ν is the kinematic viscosity, D_i is the inner manifold diameter, C_p is the specific heat and ϖ is the average flowrate. The dynamic viscosity μ can be calculated from [**28**]

$$\mu = \frac{3.3 \times 10^{-7} T^{0.7}}{1 + 0.027F} \quad (69)$$

The average flowrate ϖ is given by [**31**]

$$\varpi = \frac{NV}{120(\pi D_i^2/4)} \quad (70)$$

The natural heat convective coefficient h_n is given by [**31**]

$$h_n = 0.53 \frac{K}{D_o} (G_r P_r)^{0.25} \quad (71)$$

where G_r is the Grashof number [**31**]:

$$G_r = \frac{g\varsigma(T_m - T_{air})(0.9D_o)^3}{\nu^2} \quad (72)$$

where g is the gravitational acceleration, D_o is the outer manifold diameter and ς is the volumetric coefficient and is taken as equal to $1/T_{air}$ [**32**]. Applying the energy conservation principle to the volume, a non-linear algebraic equation in T_m is obtained. The solution to this equation gives the manifold temperature which can be substituted into equation (63) to calculate the manifold heat transfer rate:

$$T_m - T + \frac{D_o}{D_i}\frac{h_n}{h_f}(T_m - T_{air}) + \frac{D_o}{D_i}\frac{\sigma\varepsilon}{h_f}(T_m^4 - T_{air}^4) = 0 \quad (73)$$

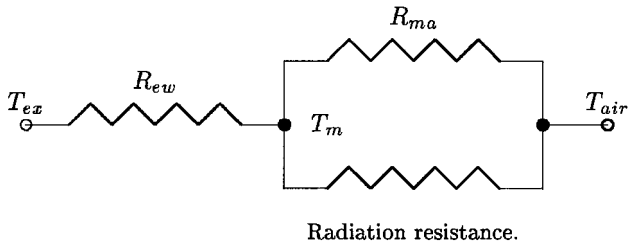

Fig. 4 Thermal resistance for the exhaust manifold

4 MODEL IMPLEMENTATION AND SIMULATION

4.1 Model implementation

The model is implemented using MATLAB/SIMULINK [**18, 19**]. The main advantage of SIMULINK is its capability for representing the entire engine model by an assemblage of interconnected blocks. It has eight variable-step solvers and six fixed-step solvers for the integration of differential equations, and hence the most suitable integration method can be chosen. Input design parameters are passed on to the blocks from an input file, but all of the operating parameters come from the block (functions) for the other components of the system. Fourth-order Runge–Kutta numerical integration is used to solve the differential equations. The integration step for the program is chosen to be one crank-angle degree. The indicated torque from the cylinder is updated at each step, and then used in the non-linear dynamic model to update the instantaneous crankshaft angular velocity. The current simulation program is executed on a 400 MHz Pentium II PC, and the simulation takes about 8 s for a 720° engine cycle at an engine speed of 950 r/min.

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4.2 Model behaviour and validation

In order to validate the transient behaviour of the engine model with experimental results, an F1L 210 DEUTZ MAG direct injection air-cooled single-cylinder diesel engine is used. The geometrical specifications for this engine are shown in Table 1.

Comparisons between the predicted and measured values of the indicated pressure and the crankshaft instantaneous angular velocity for a single cycle during the transient response are shown in Figs 5 and 6 respectively. Figure 6 also shows the predicted crankshaft instantaneous angular velocity of the model with the reciprocating and inertia variation terms removed. The measured values are taken from reference [**13**] for an engine without an exhaust manifold, hence the exhaust manifold model is removed from the simulation model for the purpose of validation. Figure 5 shows a comparison between the predicted and measured cylinder gas pressure with firing at no load and an average engine speed of 1795 r/min. The fuelling rate is 25 mm^3 per cycle. The overall agreement between the measured and predicted curves is good. The gas pressure during the exhaust stroke is slightly overestimated. The slight underestimation of the engine pressure during the combustion stroke may be due to the fact that a single-zone combustion model is used here; a slight difference in engine timing between the practical and simulated one and a minor change between the real heat transfer, heat release and the simulated one. Figure 7 shows the difference between measured and predicted cylinder gas pressure. The maximum difference is about 3.8 bar for a small duration near the top dead centre, where the measured gas pressure is about 51 bar, giving a relative percentage error of about 7.5 per cent.

Table 1 Engine geometrical specifications

Bore (d)	95 mm
Stroke	95 mm
Crank radius (r)	47.5 mm
Connecting rod length (L)	160 mm
Displacement (V_d)	673 cm^3
Compression ratio (c)	17
Piston pin offset (δ)	1.65 mm
Mass of piston and rings (M)	1.18 kg
Mass of piston (m_p)	0.98 kg
Mass of connecting rod (m_R)	0.65 kg
Intake valve opens (BTDC)	21°
Intake valve close (ABDC)	62°
Exhaust valve opens (BBDC)	62°
Exhaust valve close (ATDC)	21°
Crankshaft moment of inertia (J_c)	0.3825 kg m^2
Dynamometer moment of inertia (J_1)	0.37 kg m^2

The slight underestimation of the gas pressure during combustion has resulted in a very minor underestimation of the instantaneous engine speed with reciprocating and inertia variation near the top dead centre (TDC). In addition, the slight underestimation of the instantaneous engine speed with reciprocating and inertia variation after the mid-exhaust stroke is due to the gas pressure overestimation, as shown in Fig. 6. Other reasons for the error include inaccuracies in the engine model parameters and errors in modelling friction behaviour. The overall agreement between the measured and predicted curves is good. When the simulation is performed for

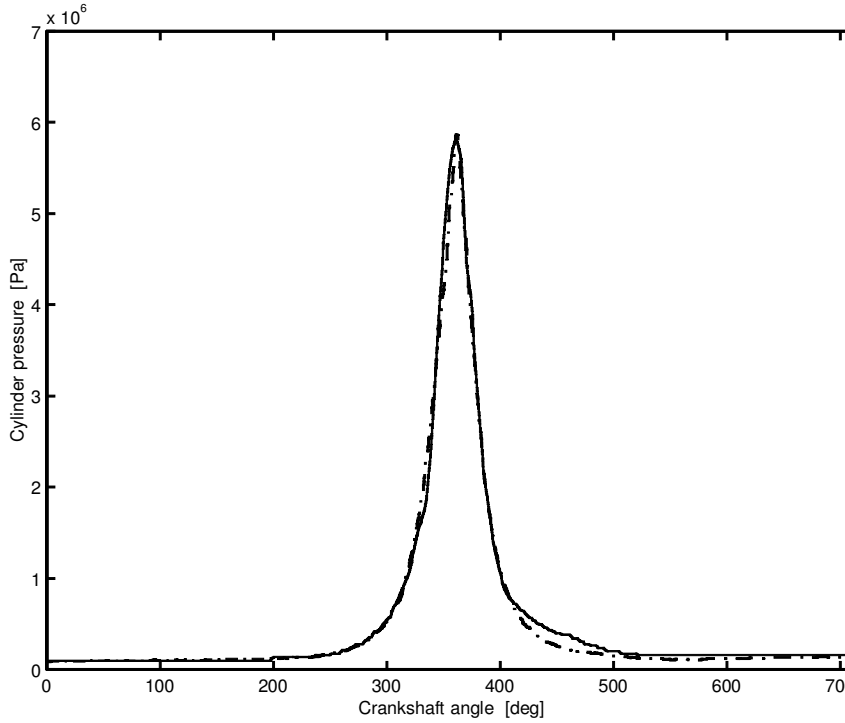

Fig. 5 Comparison between predicted (——) and measured (– – –) cylinder gas pressure under firing

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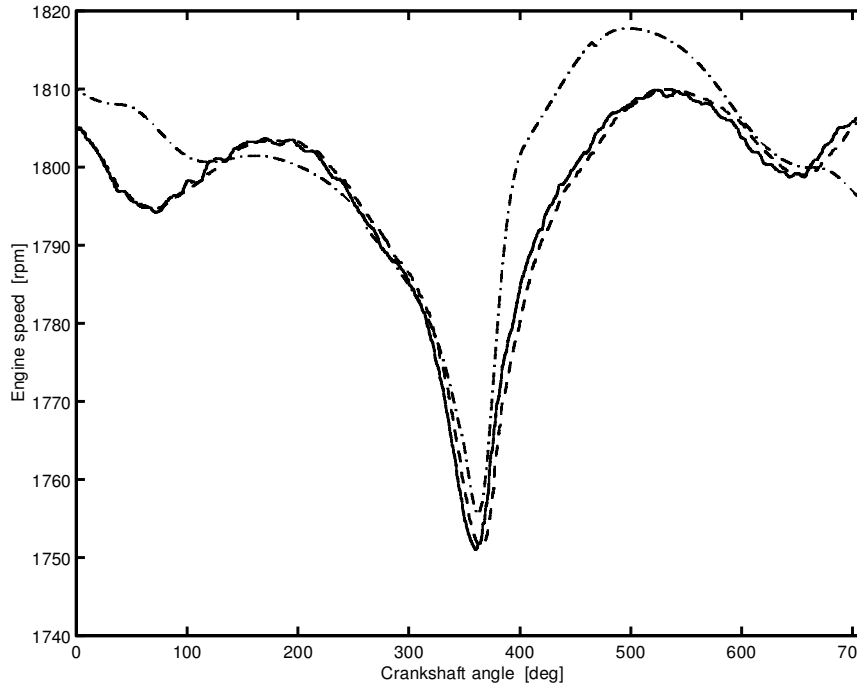

Fig. 6 Comparison between predicted with reciprocating and inertia variation (——), predicted without reciprocating and inertia variation (– · – –) and measured (– – –) instantaneous crankshaft angular velocity under firing

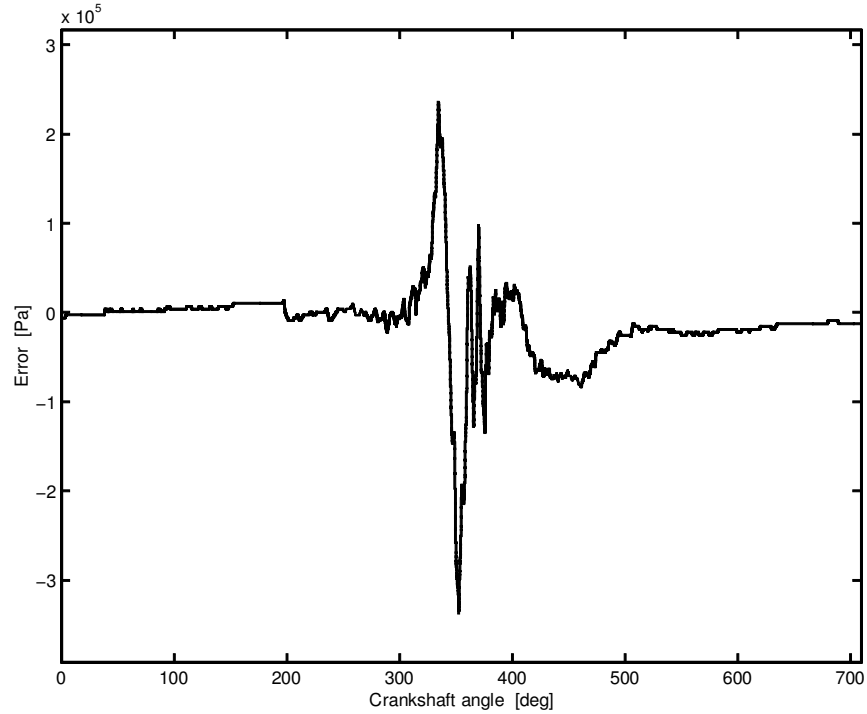

Fig. 7 Error between measured and predicted pressure

the model without the reciprocating and inertia variation terms, the speed profile has overestimation except at the end of the intake stroke, the start of the compression stroke and after the middle of the exhaust stroke. This demonstrates the importance of the incorporation of the reciprocating and inertia variation terms. The difference is more marked at high engine speed owing to the dependence of $\dot{\omega}_1$ on the ω_1^2 term [equation (6)].

The normalized rate of heat release is shown in Fig. 8. The high left peak in the curve represents the premixed

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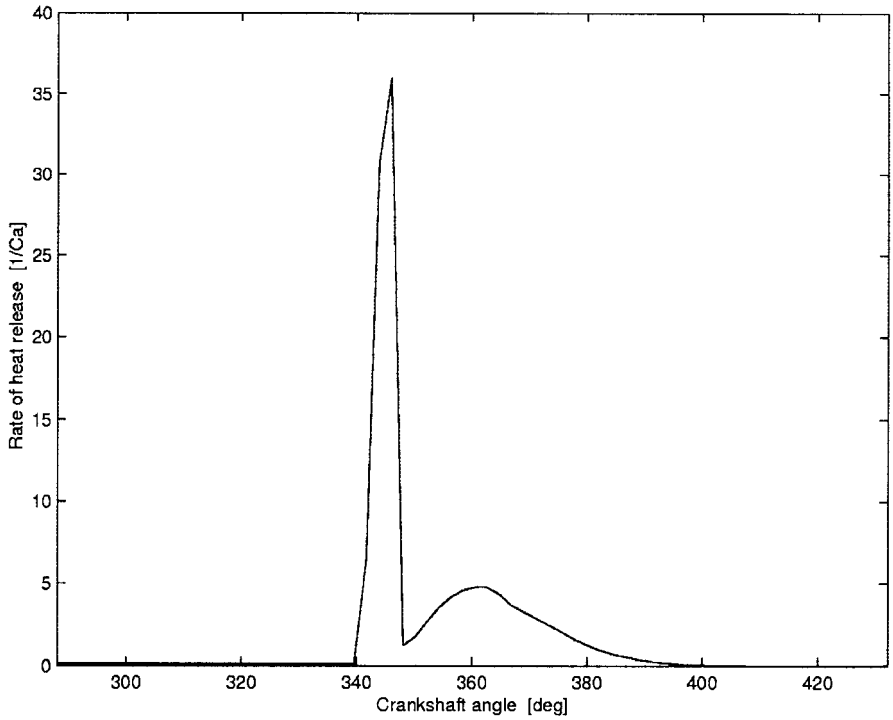

Fig. 8 Normalized heat release at 25 mm^3 of fuel per cycle

phase. Combustion of fuel mixed with air to within the flammability limits during the ignition delay period occurs rapidly in a few crank angle degrees. When this burning mixture is added to the fuel it burns during this phase, and a high heat release rate is characteristic of this phase. The low right peak is the diffusion phase; once the fuel and air premixed during the ignition delay have been consumed, the burning rate is controlled by the mixture available for burning. The heat released decreases as this phase progresses as seen in Fig. 8.

The engine model simulation is tested from starting. Figure 9 shows the comparison between predicted and measured cylinder gas pressure. Figure 10 shows the good agreement between predicted and measured instantaneous engine speed, especially at high engine speed.

There are three empirical coefficients in equations (20), (26) and (29) whose values need to be assigned. The coefficient c_1 is the empirical coefficient for ring assembly friction with mixed lubrication; the variation can be approximated by a straight line on the Stribeck diagram without significant error. The coefficient c_2 is the empirical coefficient for the friction of the bearing during mixed lubrication which depends on the properties of the pair of rubbing materials. The lubrication in the valve train at lower speed is in the boundary mode and at high speed it is in the mixed mode, so the friction is slightly decreased. The expression $1 - c_3\dot{\theta}_1$ which includes the empirical coefficient c_3 takes into consideration the slight decrease of the valve train friction coefficient with engine speed. The empirical coefficients are shown in Table 2.

Figures 11 and 12 show the instantaneous friction torque obtained from equations (18) to (30). The lubrication mode of the ring assemblies is considered to be mixed lubrication near the top dead centre. The ring assembly friction torques reach the maximum value after top dead centre where the indicated torque is maximum and its value is approximately equal to twice the ring friction torque at the beginning of the combustion. The lubrication mode of the skirt is in general hydrodynamic and its friction torque is high in the mid-region of the stroke where the piston speed is maximum and it decreases with increasing oil temperature. The bearing lubrication mode is hydrodynamic except at the points where the indicated pressure is maximum, when there is mixed lubrication and the friction torque is increased substantially; its value is approximately between 2 and 5 times the hydrodynamic lubrication friction torque value. The valve train friction torque increases during the intake and exhaust strokes owing to the spring stiffness forces being increased. The auxiliaries operate smoothly and the power taken from the engine slightly decreases when the oil temperature is increased and engine speed is constant.

The rate of engine inertia is shown in Fig. 13. $\partial J(\theta_1)/\partial\theta_1$ is a bounded periodic function with an average value of zero. Therefore, although there are periods during crank rotation when the velocity term has a positive effect and other periods where it has a negative effect, the net effect for a complete rotation is zero. The physical reason for this term is that, as the inertia decreases and less work is required for motion, it appears

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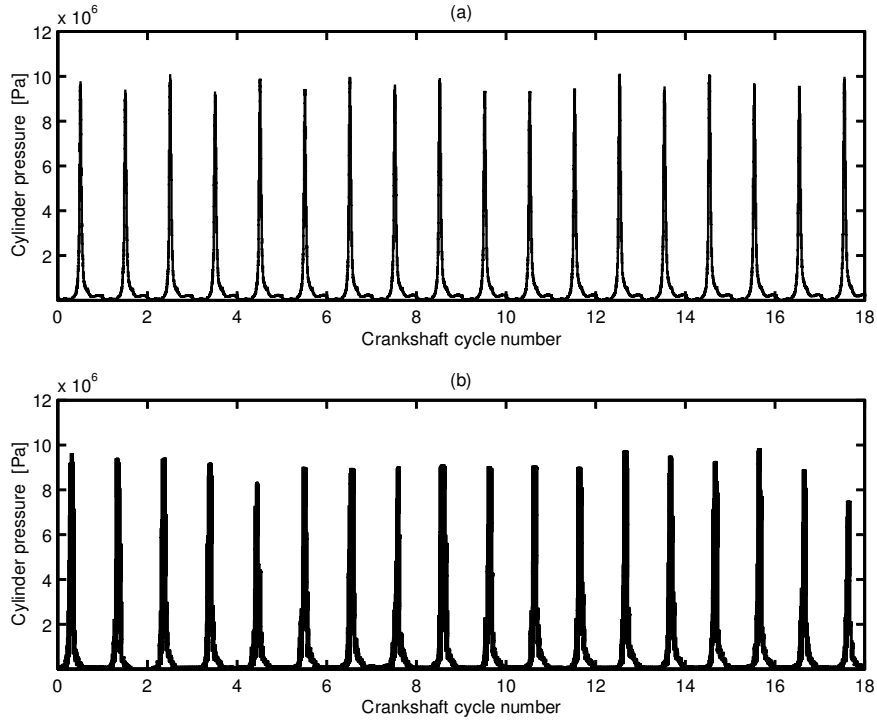

Fig. 9 (a) Predicted and (b) measured cylinder gas pressure

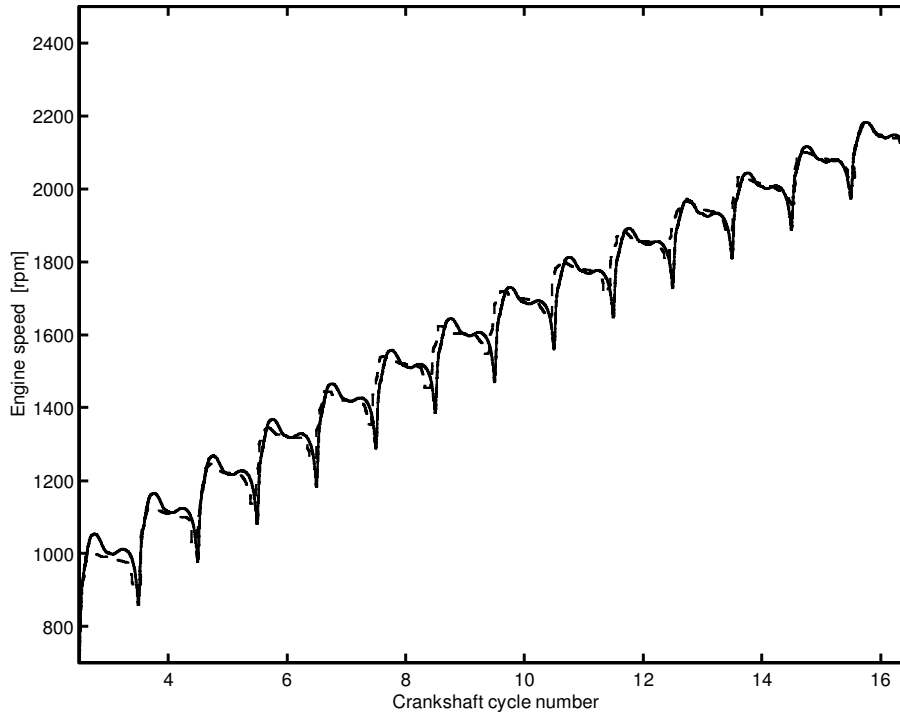

Fig. 10 Comparison between predicted (——) and measured (– – –) instantaneous crankshaft angular velocity

Table 2 Empirical coefficients

Mixed friction coefficient for rings	c_1	0.11
Mixed friction coefficient for bearings	c_2	0.002
Speed effect in valve train	c_3	0.001 27 s/rad

as if energy is added to the system. Similarly, as the inertia increases and more work is required, energy appears to be removed from the system. In the same figure, the inertia of the crank assembly is not constant, but it is a function of crank angle. Because of the attachment of the big end of the connecting rod with the crank-

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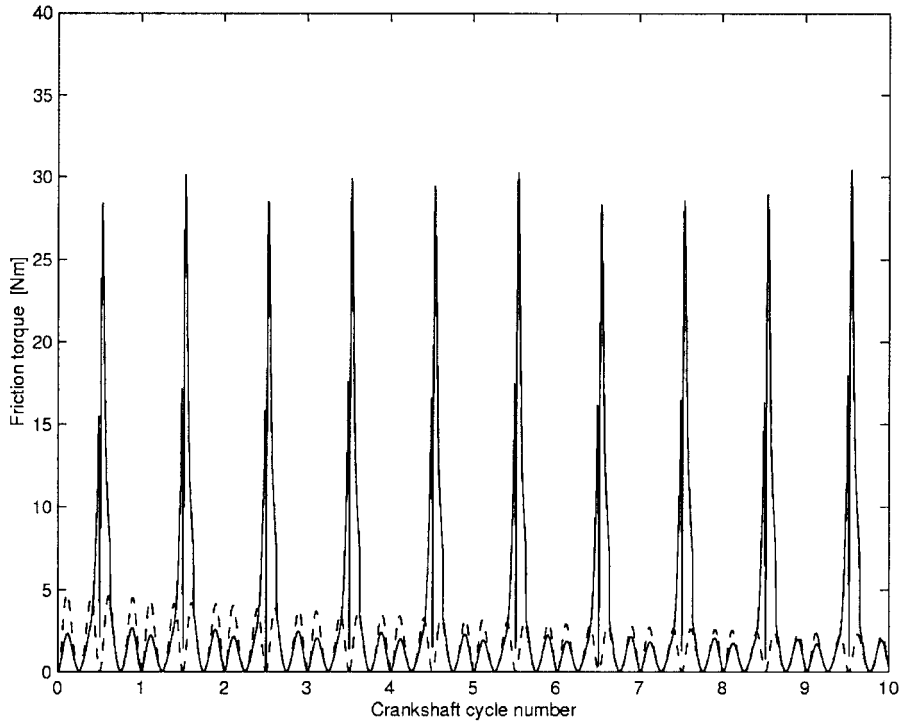

Fig. 11 Instantaneous friction torque during transient response for ring assembly (——) and skirt (– – –)

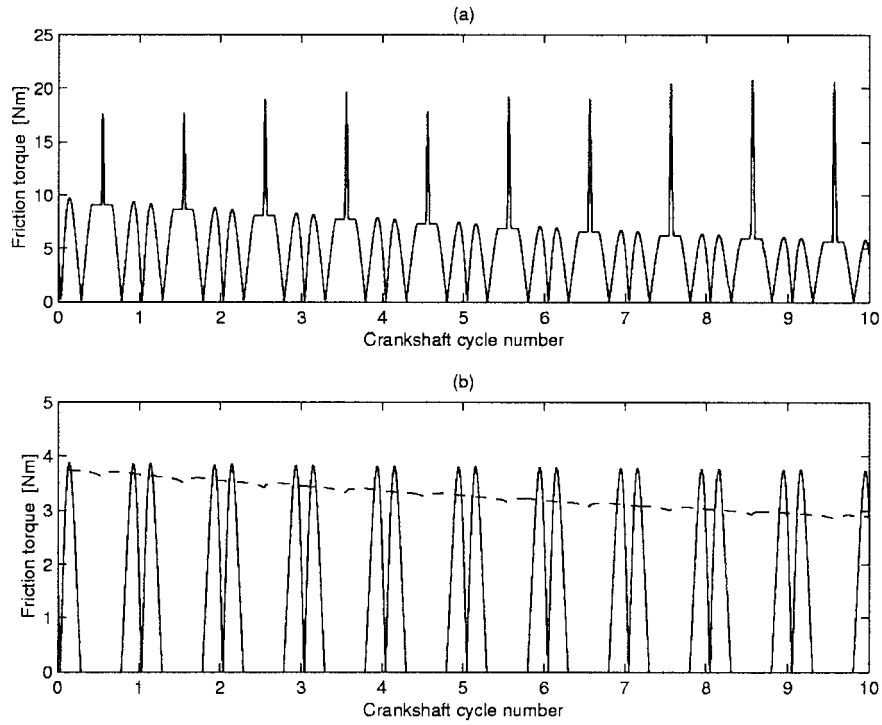

Fig. 12 Instantaneous friction torque during transient response for (a) bearings (——) and (b) valve train (——) and auxiliaries (– – –)

shaft, the maximum crank assembly inertia is near $\pi/2$ and $3\pi/2$ of the crankshaft angular position and the minimum is near 0 and π.

The instantaneous torque produced by the engine at 40 mm^3 of fuel per cycle for four engine cycles is shown in Fig. 14. The maximum torque value represents the maximum pressure in the cylinder during the combustion stroke as illustrated in the same figure. As a consequence

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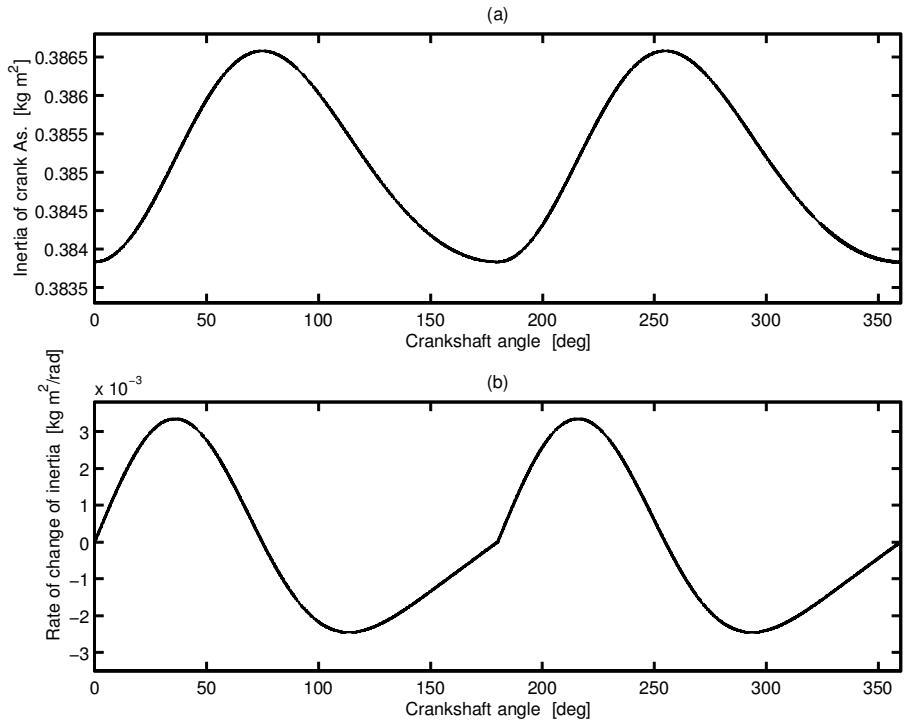

Fig. 13 (a) Inertia of the crankshaft assembly and (b) rate of the inertia for the crankshaft assembly

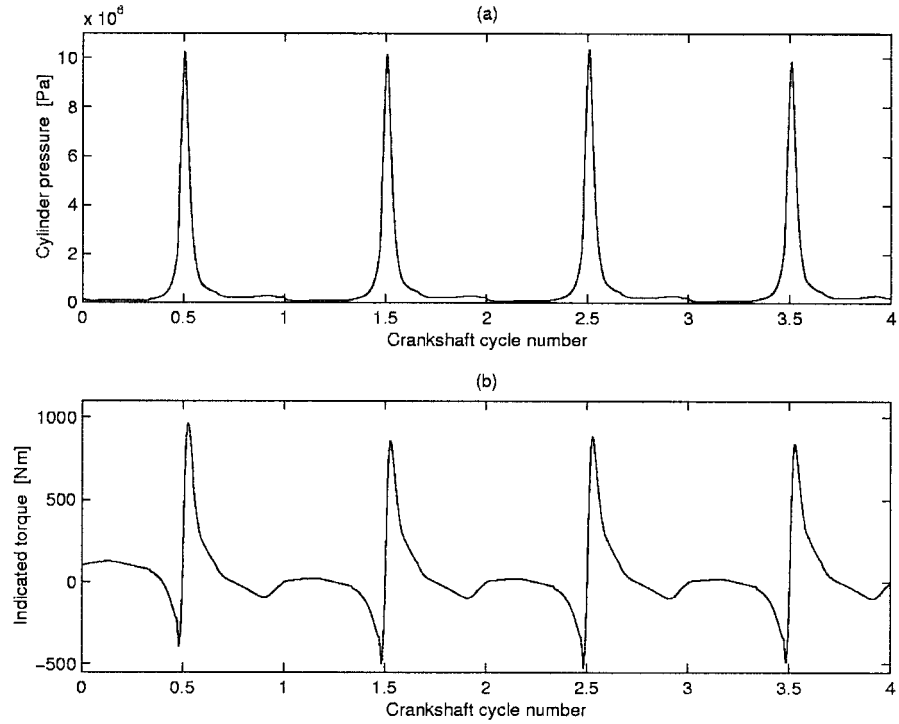

Fig. 14 (a) Cylinder pressure at 40 mm^3 of fuel per cycle and (b) indicated torque at 40 mm^3 of fuel per cycle

of the fluctuations in engine torque during the cycle, variations of the instantaneous crankshaft rotational speed occurs.

The convection and radiation heat transfer from the cylinder is shown in Fig. 15. In the vicinity of ignition at top dead centre, the magnitude of the heat transfer is high. This is due to a high gas temperature inside the combustion chamber at that moment, as seen in Fig. 16.

The rate of the equivalence ratio at 40 mm^3 of fuel per cycle is presented in Fig. 17. Since only air exists during the compression stroke, there is a zero or small equivalence ratio. Once the injection starts, the

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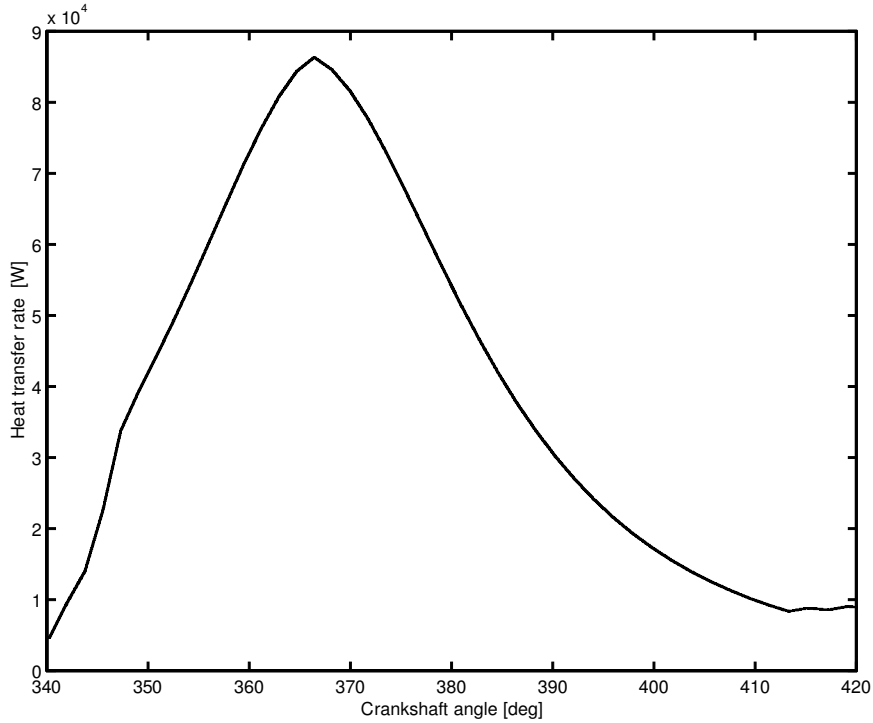

Fig. 15 Wall heat transfer rate as a function of the crankshaft angle at 40 mm^3 of fuel per cycle

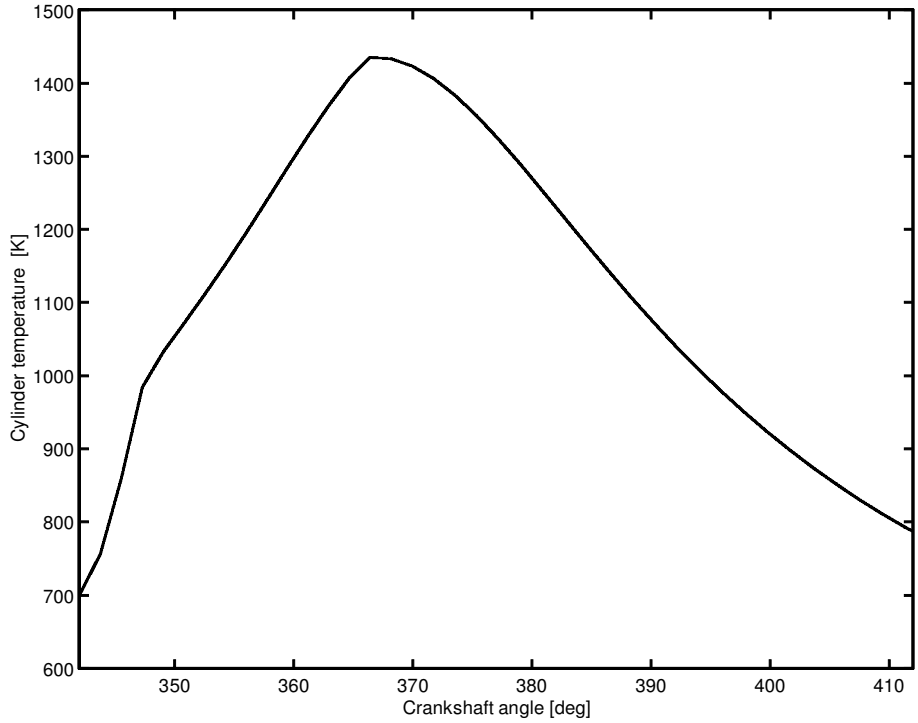

Fig. 16 Gas temperature profile as a function of the crankshaft angle at 40 mm^3 of fuel per cycle

equivalence ratio increases with a constant rate of change because the intake valves are closed and a constant fuel injection rate is assumed. After the completion of the injection the equivalence ratio is constant until all the fuel is burnt.

5 CONCLUSIONS

This paper presents a combined non-linear dynamic, instantaneous friction model in the crank angle domain with viscosity variations with temperature and thermo-

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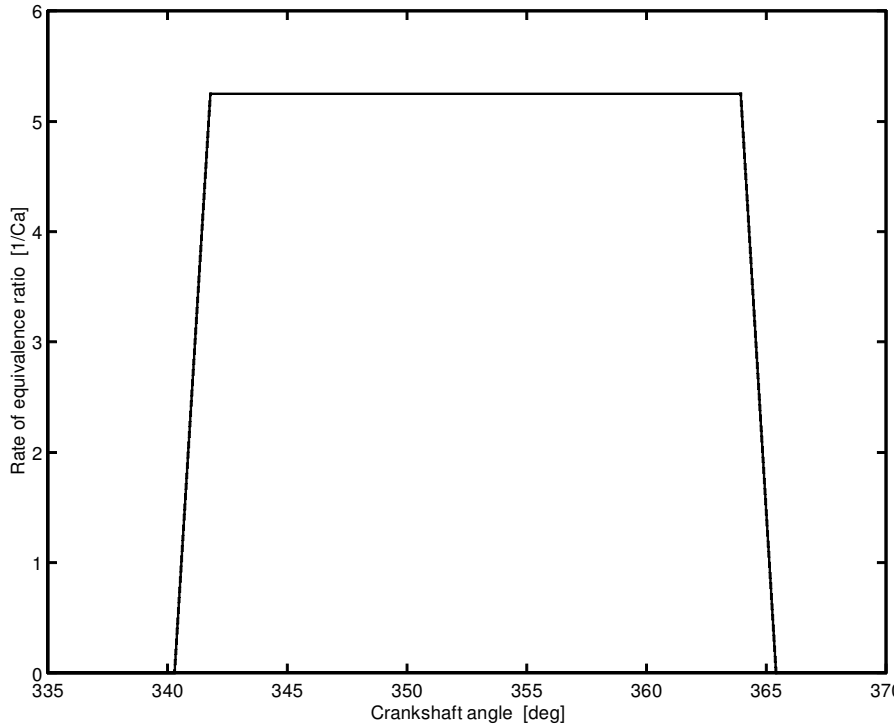

Fig. 17 Rate of the equivalence ratio for the mixture inside the cylinder at 40 mm^3 of fuel per cycle

dynamic model, for a single-cylinder diesel engine. The model is used to predict the in-cycle variation of engine states for both transient and steady state operating conditions. The model has been implemented in SIMULINK. Validation has been performed for an F1L 210 DEUTZ MAG DI air-cooled single-cylinder diesel engine. Predicted profiles of the cylinder gas pressure and the instantaneous crankshaft angular velocity through the transient and steady state cycles are in good agreement with measurements. The crank angle varying inertia with respect to the centre line of the crankshaft with piston pin offset is taken into consideration. The importance of including the effect of the inertia variation is also shown. A dynamic dynamometer model is also included, which enables a variety of engine tests to be carried out.

Accurate pressure and flowrate sensors are generally both delicate and expensive. In addition, some measurements, such as cylinder pressure, are not easily accessible. In contrast, the instantaneous crankshaft angular position and speed can be accurately measured using relatively low cost and reliable sensors. Future advances in microprocessor speeds will allow non-linear models to be used for model-based estimation, control and on-board diagnostics based on the crank angle measurements. In addition, measurement of the in-cycle speed variation can improve model-based diagnostics since oscillatory engine states are intrinsic in the model and can be related to engine failures.

The model will be used for investigating different strategies for transient fuel control and fault diagnostics. The model is being extended for multicylinder engines. Work is also ongoing to include modelling of blow-by during engine operation from cold start.

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APPENDIX

Derivation of equations for the inertia variation

Details of the derivation of inertia variation is given in this appendix. From the piston–crank mechanism, Fig. 18, the connecting rod angle can be calculated as

$$\sin\beta = \frac{\delta + r\sin(\theta_1 - \phi)}{L} \tag{74}$$

Equations (10) and (74) yield

$$\cos\beta = \sqrt{\lambda(\theta_1)} \tag{75}$$

Kinetic analysis

Considering the force equilibrium for the piston, shown in Fig. 19,

$$\begin{aligned} &\sum f_x = 0 \\ &f_{Ax} = f_{st} \end{aligned} \tag{76}$$

and

$$\begin{aligned} &\sum f_y = m_p a_{Ay} \\ &-f_{Ay} + f_i = m_p a_{Ay} \end{aligned} \tag{77}$$

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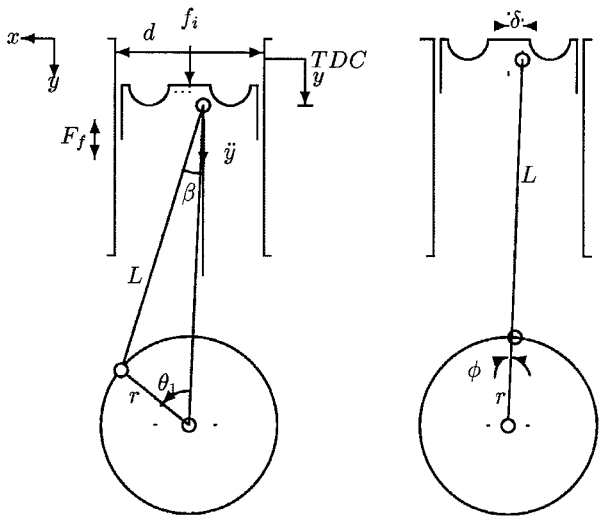

Fig. 18 Force and acceleration of the piston–crank mechanism

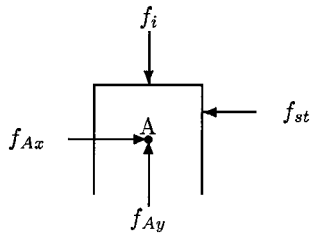

Fig. 19 Force diagram for the piston

where f_A is the reaction force from the connecting rod on the piston pin, f_{st} is the side thrust force and f_i is the force coming from the indicated pressure and the piston weight.

Considering the force and torque equilibrium for the connecting rod, shown in Fig. 20,

$$\sum f_x = m_R a_{Bx}$$
$$-f_{Cx} + f_{Ax} = m_R a_{Bx} \tag{78}$$

$$\sum f_y = m_R a_{By}$$
$$-f_{Cy} + f_{Ay} + m_R g = m_R a_{By} \tag{79}$$

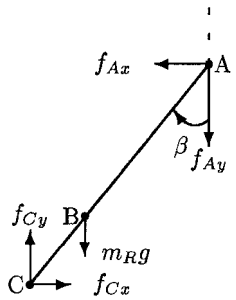

Fig. 20 Force diagram for the connecting rod

where f_C is the reaction force from the crank on the connecting rod and

$$\sum M_B = J_R \ddot{\beta}$$
$$f_{Ax}(1-b_1)L\cos\beta - f_{Ay}(1-b_1)L\sin\beta + f_{Cx}b_1 L\cos\beta - f_{Cy}b_1 L\sin\beta = J_R\ddot{\beta} \tag{80}$$

Considering the force and torque equilibrium for the crankshaft, shown in Fig. 21,

$$\sum f_x = m_c a_{cx}$$
$$-f_{Dx} + f_{Cx} = m_c a_{cx} \tag{81}$$

$$\sum f_y = m_c a_{cy}$$
$$-f_{Dy} + m_c g + f_{Cy} = m_c a_{cy} \tag{82}$$

where f_D is the reaction from force of the bearing on the crank and T is an applied torque and

$$\sum M_c = J_c \ddot{\theta}_1$$
$$T + f_{Cx}(1+b_2)r\cos\theta_1 + f_{Cy}(1+b_2)r\sin\theta_1 - f_{Dx}b_2 r\cos\theta_1 - f_{Dy}b_2 r\sin\theta_1 = J_c\ddot{\theta}_1 \tag{83}$$

Substitution of equations (76) to (79) into equation (80) gives f_{st}. Then substituting equations (81) and (82) into equation (83) yields

$$\begin{aligned} T = {} & J_c\ddot{\theta}_1 - \frac{J_R\ddot{\beta}r\cos\theta_1}{L\cos\beta} \\ & - m_p a_{py}(r\cos\theta_1\tan\beta + r\sin\theta_1) \\ & - m_R a_{Rx}(1-b_1)r\cos\theta_1 \\ & - m_R a_{Ry}(b_1 r\cos\theta_1\tan\beta + r\sin\theta_1) \\ & + m_c a_{cx} b_1 r\cos\theta_1 + m_c a_{cy} b_1 r\sin\theta_1 \\ & - m_p g(r\cos\theta_1\tan\beta + r\sin\theta_1) \\ & - m_R g(b_1 r\cos\theta_1\tan\beta + r\sin\theta_1) \\ & + m_c g b_1 r\sin\theta_1 - f_i(r\cos\theta_1\tan\beta - r\sin\theta_1) \end{aligned} \tag{84}$$

Kinematic analysis

From Fig. 22, A is connected to the piston, and in this analysis the piston is constrained to move in the

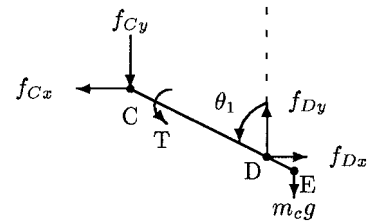

Fig. 21 Force diagram for the crankshaft

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y direction only. The velocity of the connecting rod can be calculated from Fig. 22 as

$$\dot{\beta} = -\frac{\dot{\theta}_1 r \cos\theta_1}{\cos\beta} \tag{85}$$

For the motion of the piston,

$$\begin{aligned} a_{\mathrm{p}x} &= a_{\mathrm{A}x} = 0 \\ a_{\mathrm{p}y} &= a_{\mathrm{A}y} \\ &= a_{\mathrm{C}r}\cos\theta_1 - a_{\mathrm{C}\theta_1}\sin\theta_1 + a_{\mathrm{A/C}r_1}\cos\beta \\ &\quad + a_{\mathrm{A/C}\vartheta}\sin\beta \\ &= \dot{\theta}_1^2\left[-\frac{(r\cos\theta_1)^2}{L\cos^3\beta} - r\cos\theta_1 + r\sin\theta_1\tan\beta\right] \\ &\quad - \ddot{\theta}_1(r\cos\theta_1\tan\beta + r\sin\theta_1) \end{aligned} \tag{86}$$

For the motion of the connecting rod,

$$a_{\mathrm{R}x} = \dot{\theta}_1^2(1-b_1)r\sin\theta_1 - \ddot{\theta}_1(1-b_1)r\cos\theta_1 \tag{87}$$

and

$$\begin{aligned} a_{\mathrm{R}y} &= \dot{\theta}_1^2\left[-\frac{(r\cos\theta_1)^2}{L\cos^3\beta} - r\cos\theta_1 + b_1 r\sin\theta_1\tan\beta\right] \\ &\quad - \ddot{\theta}_1(b_1 r\cos\theta_1\tan\beta + r\sin\theta_1) \end{aligned} \tag{88}$$

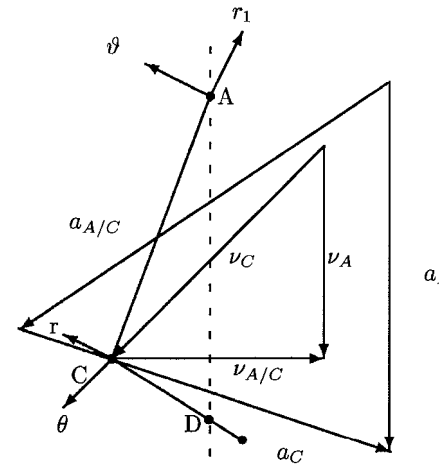

Fig. 22 Velocity and acceleration of the piston–crank mechanism

For the motion of the crankshaft,

$$a_{\mathrm{c}x} = -\dot{\theta}_1^2 b_2 r\sin\theta_1 + \ddot{\theta}_1 b_2 r\cos\theta_1 \tag{89}$$

$$a_{\mathrm{c}y} = -\dot{\theta}_1^2 b_2 r\cos\theta_1 + \ddot{\theta}_1 b_2 r\sin\theta_1 \tag{90}$$

After substituting kinematic equations (85) to (90) into equation (84), rearranging yields

$$\begin{aligned} \sum T &= J_{\mathrm{c}} + m_{\mathrm{c}}(b_2 r)^2 + J_{\mathrm{R}}\left[\left(\frac{r}{L}\right)^2\frac{1}{\lambda(\theta_1)}\cos^2\theta_1\right] + m_{\mathrm{p}}r^2\left[\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]^2 \\ &\quad + m_{\mathrm{R}}r^2(1-b_1)^2\cos^2\theta_1 + m_{\mathrm{R}}r^2\left[b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]^2\ddot{\theta}_1 \\ &\quad + \left(\frac{1}{2}\dot{\theta}_1^2\right)2J_{\mathrm{R}}\left[\left(\frac{r}{L}\right)^3\frac{\sqrt{1-\lambda(\theta_1)}}{\lambda(\theta_1)^2}\cos^3\theta_1 - \left(\frac{r}{L}\right)^2\frac{1}{\lambda(\theta_1)}\cos\theta_1\sin\theta_1\right] - 2m_{\mathrm{R}}(1-b_1)^2r^2\cos\theta_1\sin\theta_1 \\ &\quad - m_{\mathrm{R}}r^2\left[b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 - \sin\theta_1\right]\left[-\left(\frac{rb_1}{L\sqrt{\lambda(\theta_1)^3}}\right)\cos^2\theta_1 - \cos\theta_1 + b_1\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\sin\theta_1\right] \\ &\quad - m_{\mathrm{p}}r^2\left[\sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\cos\theta_1 + \sin\theta_1\right]\left[\left(\frac{r}{L\sqrt{\lambda(\theta_1)^3}}\right)\cos^2\theta_1 - \cos\theta_1 + \sqrt{\frac{1-\lambda(\theta_1)}{\lambda(\theta_1)}}\sin\theta_1\right] \end{aligned} \tag{91}$$

By comparing equation (6) with equation (91), equations (32) and (33) are obtained.

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